

CSE160 Homework 1

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Part 1

1. Normalize vector $\vec{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. Round to 2 decimal places.

$$||\vec{v}|| = \sqrt{1 + 2^2 + 3^2} \quad (1)$$

$$||\vec{v}|| = \sqrt{1 + 4 + 9} = \sqrt{14} \quad (2)$$

$$\frac{\vec{v}}{||\vec{v}||} = \begin{bmatrix} 0.27 \\ 0.53 \\ 0.80 \end{bmatrix} \quad (3)$$

2. Simplify and Find the Length of the Vector

$$3 * [1, 1] + 2 * [-1, 1]$$

$$3 * [1, 1] + 2 * [-1, 1] \quad (4)$$

$$= [3, 3] + [-2, 2] \quad (5)$$

$$= [1, 5] \quad (6)$$

And the length:

$$|[1, 5]| = \sqrt{1 + 5^2} \quad (7)$$

$$= \sqrt{26} \quad (8)$$

$$\approx 5.099 \quad (9)$$

3. Calculate the Cross Product

(a) $[0, 1, 1] \times [1, 1, 0]$

$$\det \begin{bmatrix} i & j & k \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \quad (10)$$

$$i(1)(0) + j(1)(1) + k(0)(1) - i(1)(1) - j(0)(0) - k(1)(1) = \quad (11)$$

$$j - i - k = j - k = [-1, 1, -1] \quad (12)$$

(b) $[2, 3, 4] \times [1, 0, 0]$

$$\det \begin{bmatrix} i & j & k \\ 2 & 3 & 4 \\ 1 & 0 & 0 \end{bmatrix} \quad (13)$$

$$= i(3)(0) + j(4)(1) + k(2)(0) - i(4)(0) - j(2)(0) - k(3)(1) \quad (14)$$

$$= 4j - 3k = [0, 4, -3] \quad (15)$$

(c) $[0, 3, 4] \times [2, 2, 2]$

$$\det \begin{bmatrix} i & j & k \\ 0 & 3 & 4 \\ 2 & 2 & 2 \end{bmatrix} = \quad (16)$$

$$i(3)(2) + j(4)(2) + k(0)(2) - i(4)(2) - j(0)(2) - k(3)(2) \quad (17)$$

$$= 6i + 8j - 8i - 6k \quad (18)$$

$$= -2i + 8j - 6k = [-2, 8, -6] \quad (19)$$

4. Calculate the Dot Product

(a) $[1, 0, 1] \cdot [0, 1, 1]$

$$1 \cdot 0 + 0 \cdot 1 + 1 \cdot 1 \quad (20)$$

$$= 1 \quad (21)$$

(b) $[0, 3, 4] \cdot [1, 0, 0]$

$$0 \cdot 1 + 3 \cdot 0 + 4 \cdot 0 \quad (22)$$

$$= 0 \quad (23)$$

(c) $[2, 3, 4] \cdot [6, 4, 3]$

$$2 \cdot 6 + 3 \cdot 4 + 4 \cdot 3 \quad (24)$$

$$= 12 + 12 + 12 = 36 \quad (25)$$

5. Consider a triangle formed by connecting the three points $p_1 = (0,0,0)$, $p_2 = (1,0,0)$ and $p_3 = (1,1,1)$.

First we find $p_2 - p_1$ and $p_3 - p_1$

$$p_2 - p_1 = (1 - 0, 0 - 0, 0 - 0) = (1, 0, 0) \quad (26)$$

$$p_3 - p_1 = (1 - 0, 1 - 0, 1 - 0) = (1, 1, 1) \quad (27)$$

Here, since $p_1 = \vec{0}$ we don't technically have to subtract, but a general method would subtract

Then we use the cross product.

$$p_2 \times p_3 = (0 * 1 - 0 * 1, 0 * 1 - 1 * 1, 1 * 1 - 0 * 1) \quad (28)$$

$$= (0, -1, 1) \quad (29)$$

Now, the cross product is also known to be perpendicular to its factors, meaning that $(0, -1, 1)$ has a positive z direction and is perpendicular to the triangle.

To find the area we simply divide the length of the cross product by 2 since the cross product is the area of a parallelogram.

$$|(0, -1, 1)| = \sqrt{1 + 1} = \sqrt{2} \quad (30)$$

$$A = \frac{\sqrt{2}}{2} \quad (31)$$

6. Calculate the Matrix

(a)

$$\begin{bmatrix} 1 & 2 & 5 \\ -1 & -1 & 1 \\ 4 & 4 & -2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad (32)$$

$$= \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} + \begin{bmatrix} 4 \\ -2 \\ 8 \end{bmatrix} + \begin{bmatrix} 15 \\ 3 \\ -6 \end{bmatrix} \quad (33)$$

$$= \begin{bmatrix} 20 \\ 0 \\ 6 \end{bmatrix} \quad (34)$$

(b)

$$\begin{bmatrix} 2 & 0 & 3 \\ 0 & -1 & 2 \\ 3 & 2 & -2 \end{bmatrix} \cdot \begin{bmatrix} -1 & -4 & 1 \\ 1 & -1 & 4 \\ 0 & 0 & 5 \end{bmatrix} \quad (35)$$

$$= \begin{bmatrix} -2 & -8 & 17 \\ -1 & 1 & 6 \\ -1 & -14 & 1 \end{bmatrix} \quad (36)$$

7. Consider Two lines $y = 4/3x - 1$ and $y = 0$

(a) Find the intersection of the two lines

$$0 = 4/3x - 1 \quad (37)$$

$$1 = 4/3x \quad (38)$$

$$3/4 = x \quad (39)$$

$$y = 4/3(3/4) - 1 \quad (40)$$

$$y = 1 - 1 \quad (41)$$

$$y = 0 \quad (42)$$

The intersection is $(0.75, 0)$.

(b) Are these lines Perpendicular

No.

We can directly calculate this by taking the dot product of the direction of each line. This works since $\cos(90^\circ) = 0$. If the result is nonzero then the lines must not be perpendicular.

To get the directions we take rise over run.

$$d_0 = (3, 4) \quad (43)$$

$$d_1 = (1, 0) \quad (44)$$

$$d_0 \cdot d_1 = (3 * 1 + 4 * 0) = 3 \quad (45)$$

Part 2

1. Define OpenGL, OpenGL ES and WebGL. Describe their relationship.

OpenGL is a standard API for communicating to graphics drivers and their associated graphics processing units. The main characteristics of OpenGL is being fairly high level in terms of API and being an open standard meaning it is not proprietary to any specific platform or device.

OpenGL ES is a version of the OpenGL standard geared towards mobile devices and embedded systems, hence the name OpenGL ES (Embedded Systems).

WebGL is an API provided by browsers to javascript websites in order for web applications to utilize GPU acceleration. It is based on OpenGL ES.

Each of these are related in the following ways. OpenGL is the oldest. OpenGL ES was based on an OpenGL version, meaning OpenGL ES 1.1 was based on OpenGL 1.5, and OpenGL ES 2.0 was based on OpenGL 2.0. WebGL is based on OpenGL ES, and tracks along with the ES standard.

2. What is the difference in the software architecture of a webpage and a webpage using WebGL. Describe all components involved in the two cases.

A standard webpage uses 3 languages. First html defines the main structure of the page, defining each element.

Then CSS is used to layout the page, setting colors, styling and how html elements should be organized.

Javascript is then run in an event loop which provides interactivity in the browser without needing a full roundtrip request.

A WebGL webpage simply adds a connection to OpenGL ES which provides an api in javascript.

These translate to the following software browser components, an html rendering engine that handles CSS, a javascript runtime, and an OpenGL ES context, which is provided by the driver.

3. In HTML, what is the `<canvas>` tag?

The Canvas tag provides a buffer and html element which javascript can render to. It can be written to via cpu side instructions, and in the context of WebGL it is a buffer which can be rendered to and displayed on the webpage.

4. In Javascript, what line of code one writes to retrieve a 2D rendering context?

Assuming you already have the canvas, meaning you did something like

```
var canvas = document.getElementById("name");
```

According to the Mozilla developer documentation to get a webgl context you do:

```
var gl = canvas.getContext("webgl");
```

According to the textbook the specific string used to get the context varies between browsers so you would normally replace it with a function.